Liqin Ye

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Education

- **Georgia Institute of Technology** — Atlanta, GA, USA
  Ph.D. in Machine Learning — *Aug 2024 - Jun 2029 (Expected)*
  M.S. in Computer Science — *Aug 2023 - Dec 2025*
  - **GPA**: 3.85/4.00
  - **Research Interests**: LLM agent, LLM reasoning, LLM in Finance.
  - **Advisor**: Dr. Chao Zhang and Dr. Sudheer Chava
- **University of California, Irvine** — Irvine, CA, USA
  B.S. in Computer Science (ICS Honor Student) — *Aug 2019 - Jun 2023*
  - **GPA**: 3.95/4.00 (Magna Cum Laude)
  - **Core Courses**: Deep Generative Models, Deep Learning, Graph Algorithms, Formal Language & Automata
  - **Advisor:** Dr. Stephan Mandt

Professional Experience

- **Rufus Post-training Team, Amazon** — Palo Alto, CA, USA
  Applied Scientist Intern, Mentor: *Rongzhi Zhang*, Manager: *Qingyu Yin* — *May 2026 - Present*
  Topic: Multi-step Agentic RL Training for LLM Forecast Agent

Research Project

- **Multi-Step Reinforcement Learning for LLM Forecasting Agent** — *May 2026 - Present*
  - Formulated real-world event forecasting as sequential posterior approximation, training an LLM agent to continuously revise probabilistic beliefs as new evidence arrives rather than make one-shot predictions
  - Designed a leak-free temporal POMDP over a daily-sliced news corpus, enabling date-restricted SEARCH/SCRAPE/CODE interactions and a structured rolling belief notebook for long-horizon evidence tracking
  - Built an end-to-end data pipeline to curate resolving real-world questions using temporal contamination controls, hybrid BM25/dense retrieval, news-dynamics filtering, and LLM-based corpus solvability checks
  - Improved GLM-4.5-Air (106B, 12B active) forecast accuracy from 39.5% to **50.0%** and Brier score from 0.802 to **0.625**
- **Evolutionary Task Discovery for LLM Reasoning** — *Jun 2025 - Jan 2026*
  - Formulated reasoning data generation as directed search over a manifold of algorithmic skill and complexity
  - Designed two evolutionary operators, Attribute Mutation for complexity scaling and Skill Crossover for compositional logic discovery, with a dynamic Zone of Proximal Development filter to ensure learnability
  - Achieved **+5.7%** on code and **+6.4%** on math benchmarks over Qwen3-4B Thinking, with up to **+8.2%** on competition-level AIME problems; consistent gains in both pass@1 and pass@8 confirm no diversity collapse.
  - Validated effectiveness and scalability on Base models (Qwen3-4/8B) and Instruct models (LLaMA-3-3/8B)
- **Precise Attribute Intensity Control in LLMs** — *Nov 2024 - May 2025*
  - Developed a test-time gradient-based representation editing method (PRE-CONTROL) that steers LLM's hidden states toward multi-dimension attribute intensity targets in closed-form to meet diverse user preferences
  - Implemented TD-λ training for a lightweight value function to approximate faithful partial sequence reward
  - Achieved **6.6-30.7%** exact target-hit success (up to **5.1x** over best baseline) on both text and code generation tasks
  - Improved Pareto frontier hypervolume from 7.54 to **12.66** while reducing **8x** computational overhead
  - Designed a controllable distillation pipeline using PRE-CONTROL, reaching higher Pareto hypervolume (**16.81** vs. 15.27) with only **15k** vs. 50k samples and **2.1** vs. 7.8 GPU hours compared to Best-of-N distillation
- **Calibrating Pre-trained Classifiers on LLM-generated Noisy Labels** — *Sep 2023 - May 2024*
  - Developed label candidate retrieval in embedding space to obtain faithful true labels for noisy samples
  - Stabilized the training of true label generative modeling via label candidates distillation algorithm
  - Designed a Simplex Label Diffusion Model to recover true labels from noisy labels via iterative denoising process
  - Improved BERT accuracy by **7.3%** on average across 4 NLP tasks and 5 LLMs annotated labels, and up to **11.7%** over raw LLM predictions and **4.5%** over the best noisy-label baseline on SemEval.

PUBLICATIONS

1. **Liqin Ye**, Yanbin Yin, Michael Galarnyk, Yuzhao Heng, ..., Chao Zhang. **Evolutionary Task Discovery: Advancing Reasoning Frontiers via Skill Composition and Complexity Scaling**. Submitted to ***NeurIPS*** 2026.
2. Rongzhi Zhang*, **Liqin Ye***, Yuzhao Heng, Xiang Chen, Tong Yu, ..., Chao Zhang. **Precise Attribute Intensity Control in Large Language Models via Targeted Representation Editing**. Preprint.
3. **Liqin Ye**, Agam Shah, Chao Zhang, Sudheer Chava. **Calibrating Pre-trained Language Classifiers on LLM-generated Noisy Labels via Iterative Refinement**. In *Proceedings of the 31st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**)*, 2025.
4. Michael Galarnyk, ..., **Liqin Ye**, ..., Sudheer Chava. **IPO-Mine: A Toolkit and Dataset for Section-Structured Analysis of Long, Multimodal IPO Documents**. In *Proceedings of the 32nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**) Dataset & Benchmark Track* 2026.
5. Agam Shah, **Liqin Ye**, ..., Wei Xu, Sudheer Chava. **Beyond the Reported Cutoff: Where Large Language Models Fall Short on Financial Knowledge**. In *Proceedings of the 2nd Conference on Language Modeling (**COLM**)*, 2025.
6. Agam Shah, ..., **Liqin Ye**, ..., Sudheer Chava. **Words That Unite The World: A Unified Framework for Deciphering Central Bank Communications Globally**. In *Proceedings of the 39th Annual Conference on Neural Information Processing Systems (**NeurIPS**) Dataset & Benchmark Track*, 2025.

HONORS & AWARDS

- **Workday AI PhD Fellowship**, Workday AI Research — *2026*
- **ICS Honor Student**, University of California, Irvine — *2023*

SKILLS

- **Programming Languages:** Python, C/C++, JavaScript, TypeScript, SQL, Scala, HTML
- **Frameworks & Libraries:** PyTorch, Slime, Verl, vLLM, SGLang, Ray, LlamaIndex, Numpy, Git, Pandas
- **Language:** Mandarin, English
- **Passions:** Snowboarding, Basketball, Calligraphy, Photography

SERVICES

- **Teaching:** MGT 8997 AI in Finance, CS 7641 Machine Learning, ICS 45C Program in C++
- **Reviewer:** NeurIPS 2026, ARR 2026, KDD 2025, NeurIPS 2024, ARR 2024, Computational Economics
- **Conference:** AI and Future of Finance (Setup, Poster Session)